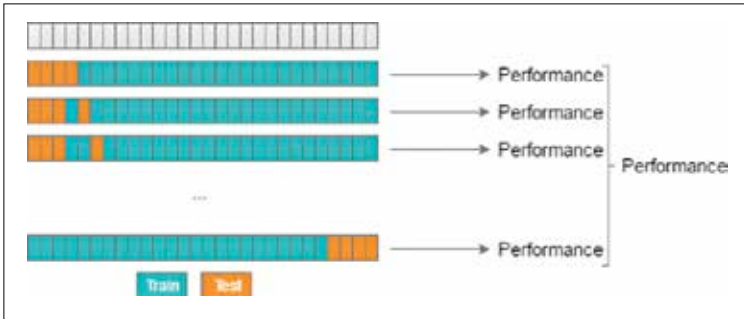




$p=4$ in Leave-P-Out Cross Validation

The above validation technique is exhaustive as it computes all ways of splitting the original dataset. We do not have to explicitly mention the number of splits.

Cross Validation is a very useful technique for assessing the effectiveness of your model, particularly in cases where you need to mitigate overfitting. It is also of use in determining the hyper parameters of your model, in the sense that which parameters will result in lowest test error. This is all you need to get started with cross validation.

Testing

The holdout method used in the validation discussed earlier is used to make the testing dataset. The testing dataset is never seen by the model during training as is the best measure to check the actual performance of the model. As stated earlier 75/25 and 80/20 split is generally used for training and testing.

This brings us to the end of the theory of Training and Validation. You can implement the above mentioned algorithms using Python and its extensive libraries available. Pandas, numPy,sciPy need to be installed before we begin implementation. If you are working in an Ubuntu shell use:

- `sudo pip install numpy scipy pandas`

Scikit-learn is an open-source library part of sciPy which has an in-built implementation of all the above mentioned algorithms. It has got content that is relevant for any level of the coder.